1)A machine has 48-bit virtual addresses and 32-bit physical addresses. Pages are 8 KB.

1. Obtain the number of pages and page frames.
   8Kb = 2^13 B

   # of pages 2^48/2^13 = 2^35

   # of page frames2^32/2^13 = 2^19

2. How many entries are needed for a single-level linear page table?

   2^48/2^13 = 2^35

2)Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 MB, 4 MB, 20 MB, 18 MB, 7 MB, 9 MB, 12 MB, and 15 MB. Which hole (in MB) is taken <u>for successive segment requests</u> of

(a) 12 MB

(b) 10 MB

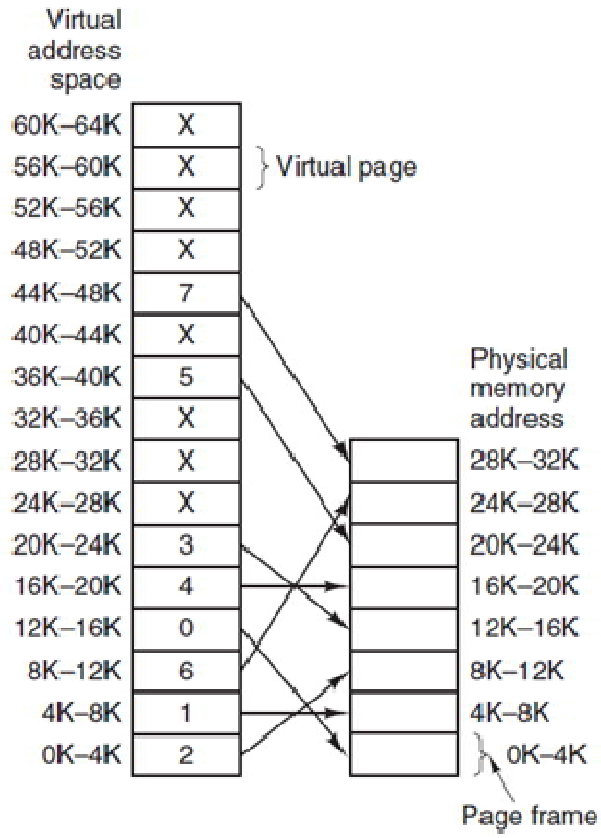
(c) 9 MB

For

	12 MB	10 MB	9 MB
first fit	20	10	18
best fit	12	10	9
worst fit	20	18	15
next fit	20	18	9

3) Using the following page table,

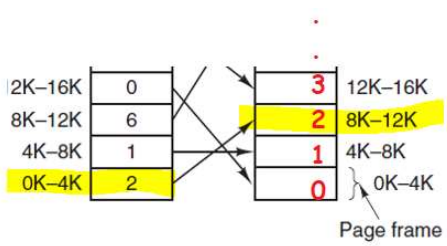
a) Obtain the page size.
   e.g.64k - 60k = 4K
b) Obtain the page frame size
   e.g. 32k-28k = 4k
c) Using direct calculation and the formula, give the physical address corresponding to each of the following virtual addresses:
   1. 20
   2. 4100
   3. 8300
   4. 53448



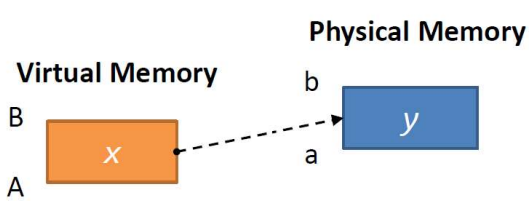
1) 20

Offset = 20 – 0 = 20

Physical address = 20 +8*1024 = 8212



Formula:



Y = x + (b-B) or y = x + (a-A) where y = Physical address

Y = x + (a-A)

A = 0, a = 8K = 8x1024, B = 4K = 4096, b = 12K = 12x1024 = 12288

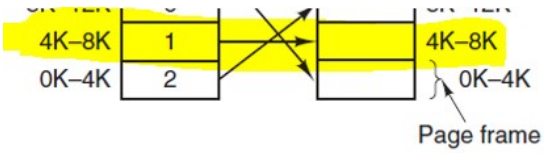
Y = 20 + (8192 -0) = 8212

Or

Y (Physical address) = x + (b-B)

Y = 20 + (12288-4096) = 8212

2) 4100:



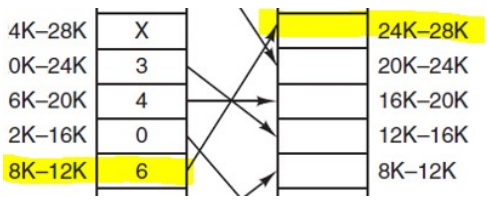
A = 4K, a = 4K

Offset = (4100-4*1024)

Physical address = (4100-4*1024) + 4*1024 = 4100

Y (Physical address) = x + (a-A) = 4100 + (4K-4K) = 4100

3) 8300 :



A = 8K, a = 24K

Offset = (8300-8*1024)

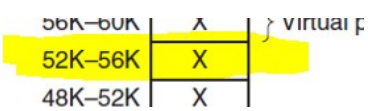
Physical address = (8300-8*1024) + 24*1024 = 24684

Y (Physical address) = x + (a-A) = 8300 + (24K-8K) = 8300+16K

Y = 8300+16x1024 = 24,684

4) 53448

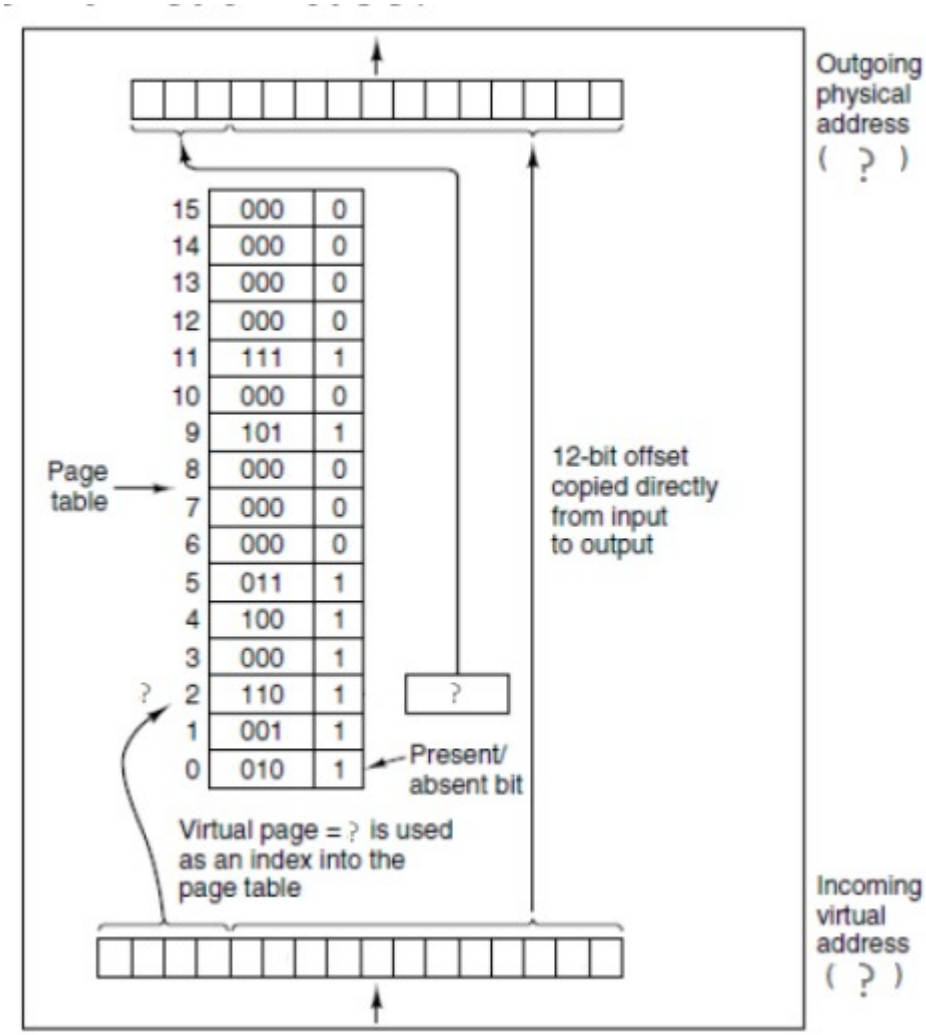
Page fault



4) Consider the following MMU with 16 4-KB pages and following virtual addresses.

a) 25012(0X61B4)
b) 45493(0XB1B5)

Obtain a physical address for each case.



Total virtual addresses = 16 x 4 x 1024 = $2^4 \times 2^2 \times 2^{10} = 2^{16}$ (Virtual address: 16 bits)

a) 0110 000110110100 -> $110_2 = 6_{10}$ Page fault
b) 1011 000110110101 -> $1011_2 = 11_{10}$ physical address $111000110110101_2 = 29109_{10}$

5) Consider the following resource allocation graph and obtain C, R , and E matrices.

C

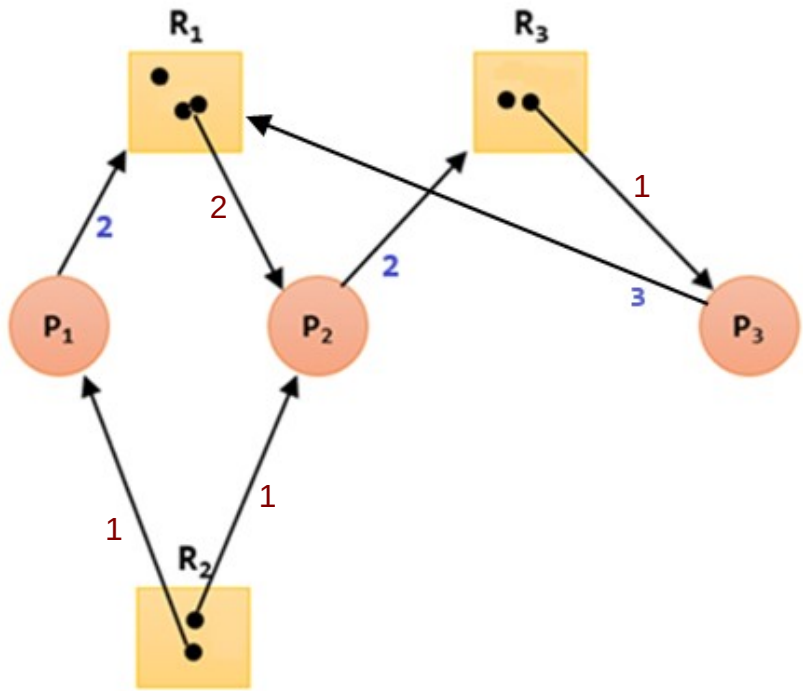
	R1	R2	R3
P1	0	1	0
P2	2	1	0
P3	0	0	1

R

	R1	R2	R3
P1	2	0	0
P2	0	0	2
P3	3	0	0

E

R1	R2	R3
3	2	2

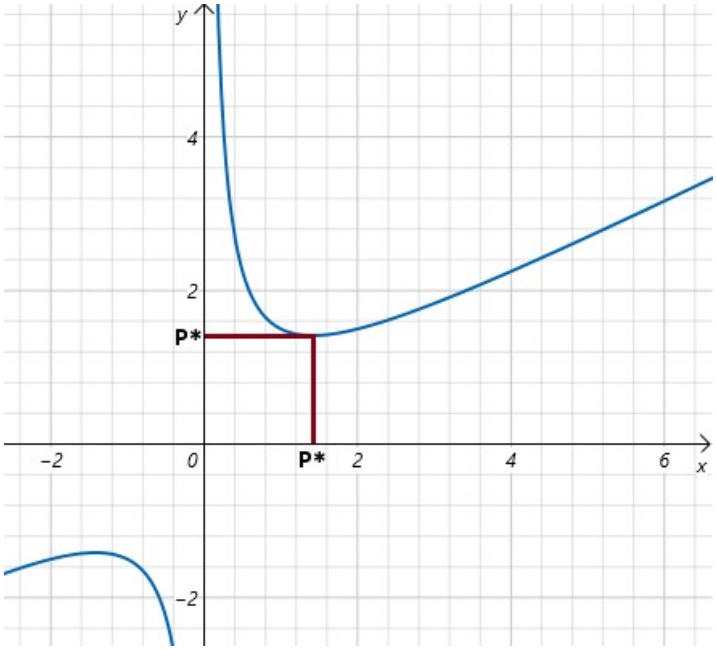


6) Using the provided formula for memory overhead associated with page size; calculate the overhead for the optimal page size.

$$\text{Over head} = \frac{se}{p^*} + \frac{p^*}{2}$$

$$p^* = \sqrt{2se}$$

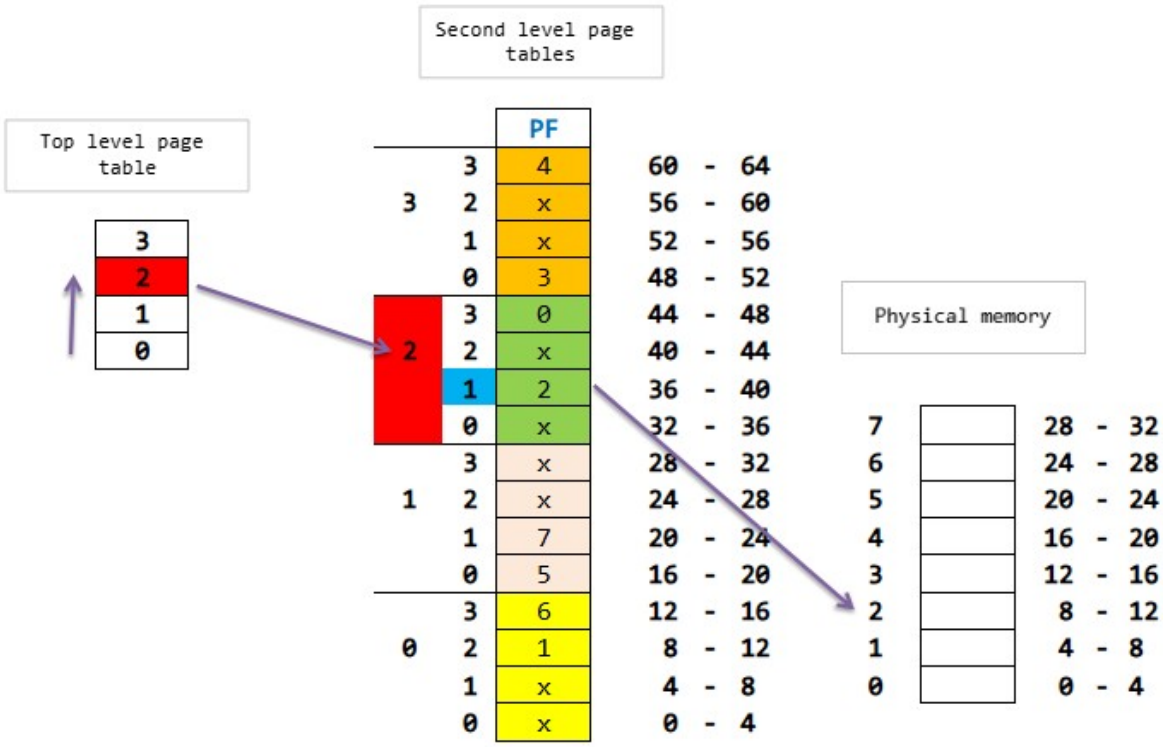
$$\text{Over head} = \frac{se}{\sqrt{2se}} + \frac{\sqrt{2se}}{2} = \sqrt{2se} = p^*$$



7) Consider the following two-level page table and obtain

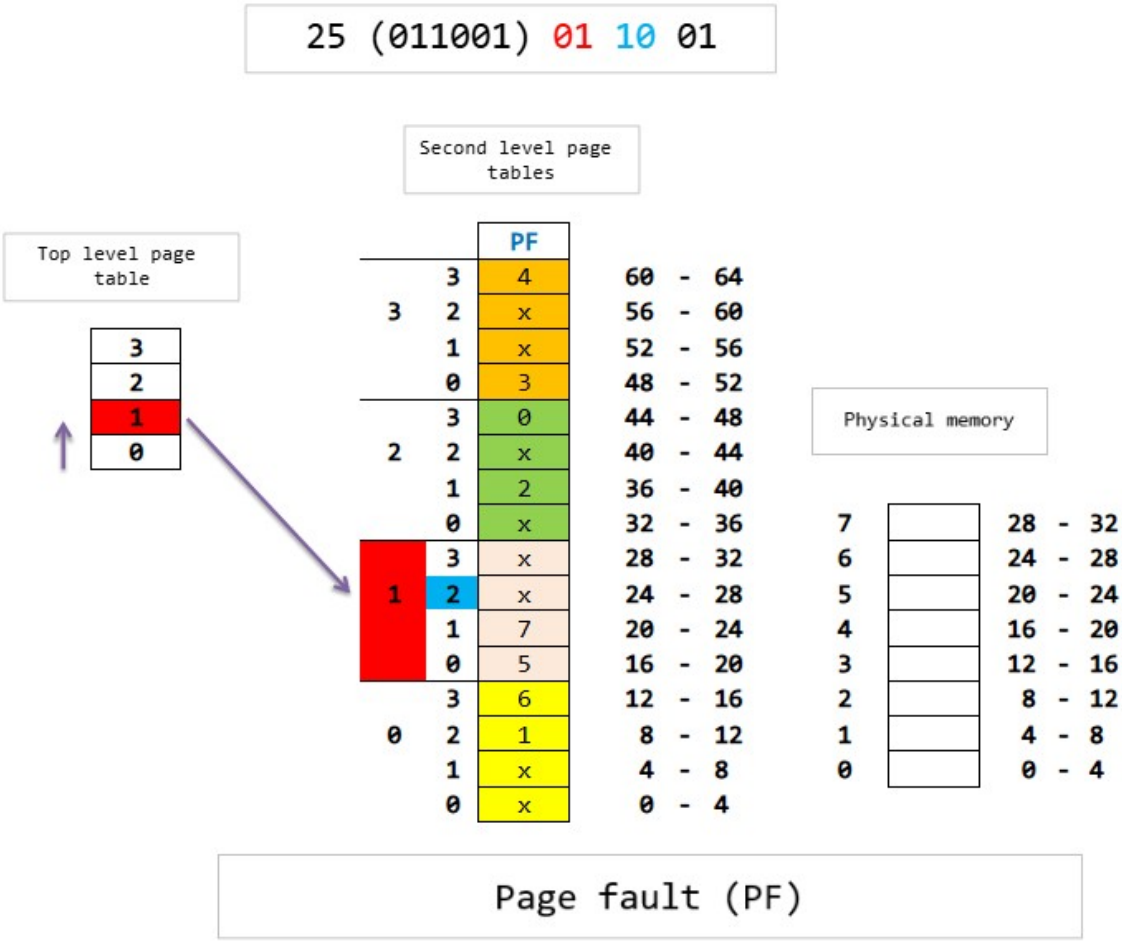
a) The size of virtual address space and physical memory. 64 and 32 bytes
b) The page and page frame size. 4 bytes
c) The physical address for each of the specified scenarios
   i. 37 (0X25)
   ii. 25 (0X19)
   iii. 62 (0X3E)

37 (100101) 10 01 01

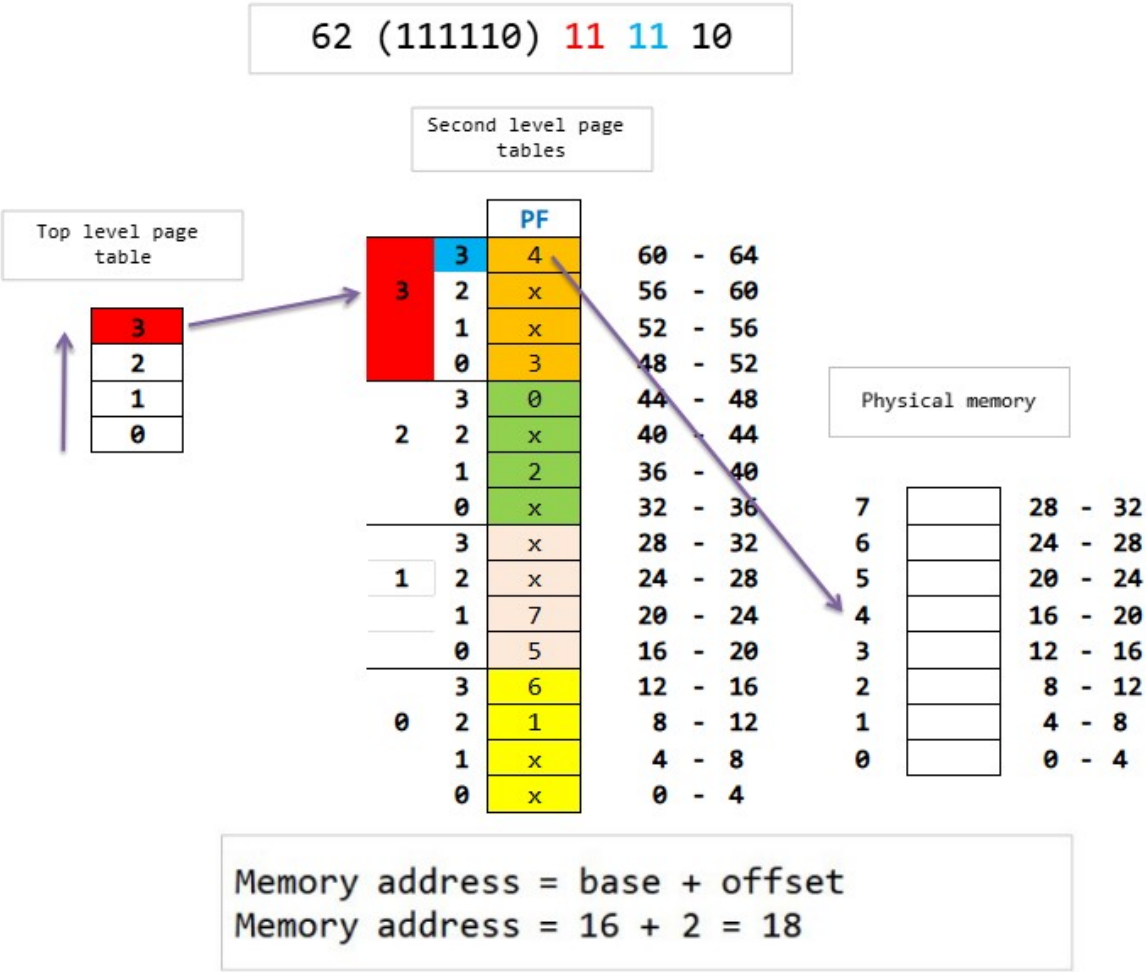


Memory address = base + offset
Memory address = 8 + 1 = 9

Top level page (index)	Second page tables (index)	Memory address
2	1	8 + 1 = 9



Top level page (index)	Second page tables (index)	Memory address
1	2	X : Page fault



Top level page (index)	Second page tables (index)	Memory address
3	3	16 + 2 = 18

8) A system has 1024 KB of memory, and the smallest block size is 32 KB. The following sequence of operations occurs:

Process A requests 200 KB.
Process B requests 50 KB.
Process A releases its memory.
Process C requests 90 KB.

Questions:

1. Show the allocation process for each request.
2. Illustrate the memory tree after each operation.
3. Show the coalescing process after Process A releases memory.

```
1024 KB
├── 512 KB
│   ├── 256 KB (Allocated to Process A)
│   └── 256 KB (Free)
└── 512 KB (Free)
```

```
1024 KB
├── 512 KB
│   ├── 256 KB (Allocated to Process A)
│   └── 256 KB
│       ├── 128 KB
│       │   ├── 64 KB (Allocated to Process B)
│       │   └── 64 KB (Free)
│       └── 128 KB (Free)
└── 512 KB (Free)
```

```
1024 KB
├── 512 KB
│   ├── 256 KB (Free)
│   └── 256 KB
│       ├── 128 KB
│       │   ├── 64 KB (Allocated to Process B)
│       │   └── 64 KB (Free)
│       └── 128 KB (Free)
└── 512 KB (Free)
```

```
1024 KB
├── 512 KB
│   ├── 256 KB (Free)
│   └── 256 KB
│       ├── 128 KB (Allocated to Process C)
│       └── 128 KB
│           ├── 64 KB (Allocated to Process B)
│           └── 64 KB (Free)
└── 512 KB (Free)
```